RPMGLOBAL

AMT – Asset Benchmarking & Economic Life Optimisation

Training Manual

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding on the workshop and ask any related questions.

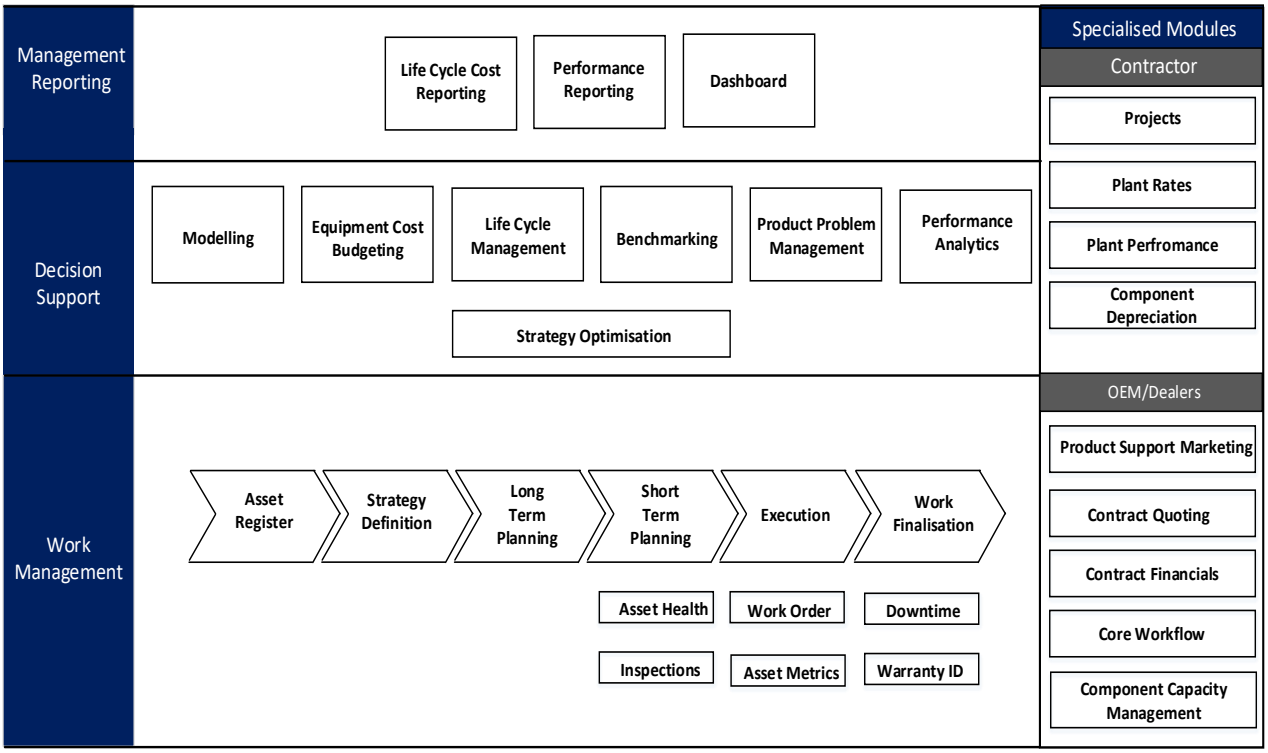
Course Outline

Prerequisites	None
Target Audience	Strategists, Long-Term Planners, Reliability Engineers, Procurement, C-Level Management
Nominal Duration	2 Hours
Course Objectives	This course aims to equip participants with the knowledge and skills to use AMT.

Learning Outcomes	At the end of the course, participants will have learned: • Asset Benchmarking ○ Run Equipment Benchmark Reports. ○ Investigate Performance drivers. • Economic Life Optimisation ○ Identify Equipment in replacement period. ○ Validate optimal Equipment Replacement points. ○ Validate Equipment's life cycle term. ○ Refine Maintenance Strategy. ○ Refine Long-Term Plan.
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below.

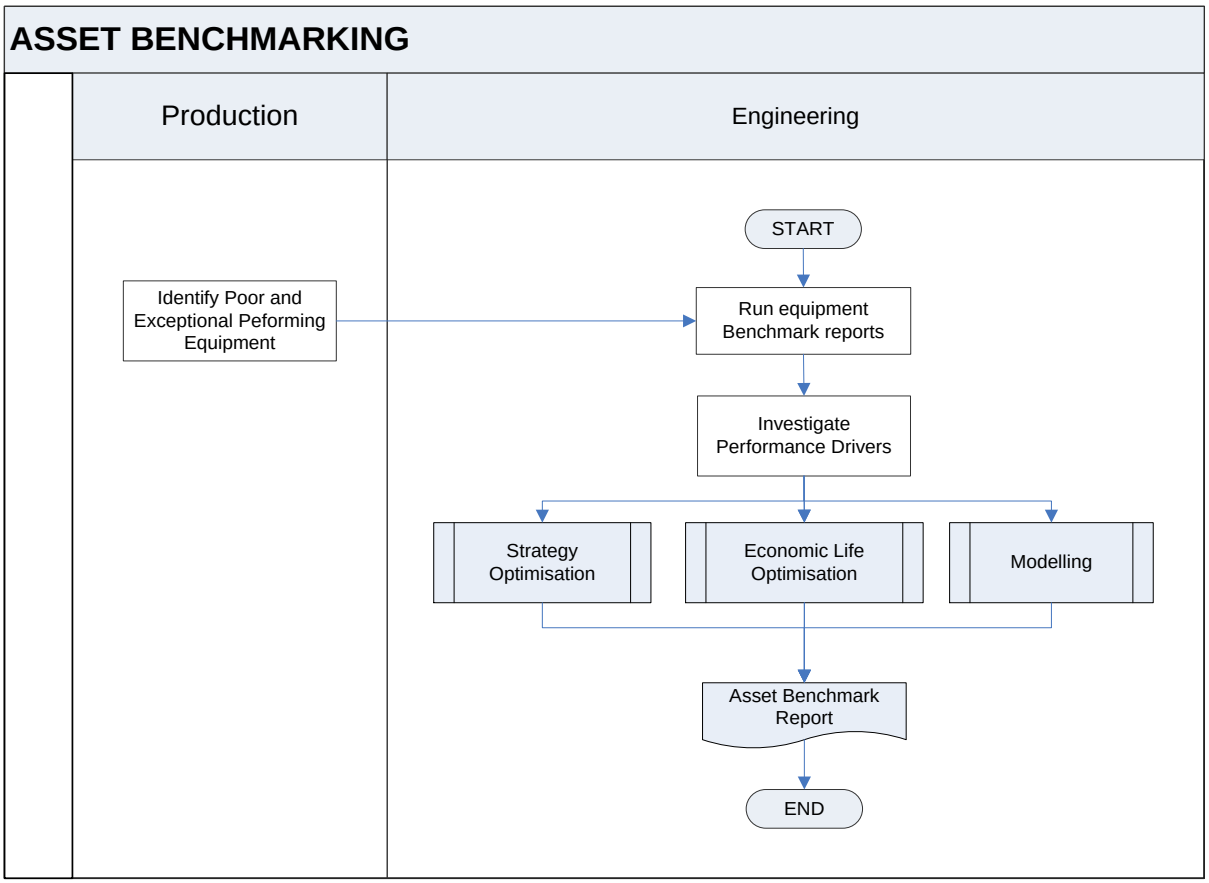


Asset Benchmarking

Overview

Asset Benchmarking is the process of comparing asset performance and maintenance strategies within a single site and/or across multiple sites.

Process Diagram



Training Workshops

1. Run Equipment Benchmark Reports

Overview

AMT has a suite of standard industry-recognised Benchmarking reports. These reports are run at an asset level or at the level of specific components, to gather a clear understanding of an asset's performance by identifying the assets or components which are costing or performing higher or lower than what is ideal.



Your ability to run benchmarking reports will depend on whether you have access to the relevant Branch or Site.

Procedure – Run Equipment Benchmark Reports

Perform the following steps to run Equipment benchmark reports.

1. Search for **Cost per UOM** report.



*Alternatively, from the **My Functions** hierarchy select **Reporting & Analyses** > **Equipment Evaluation** > **Cost per UOM.***

2. The **Cost per UOM** function has four options:

   - Cost per UOM - by Date
   - Cost per UOM - by Usage
   - Cost per UOM - Comparison
   - Cost per UOM per interval

   To perform benchmarking based on asset usage, select **Cost per UOM - by Usage**.



• Not all assets will require benchmarking at an asset level.

• In Cost per UOM, benchmarking occurs at a total asset level such as cost per hour or cost per tonnes.

3. On the **Cost per UOM by Usage - Report Parameters** dialog, in the **Data Selector** section, set the date range for the report, either **Entire Plan Term** or **Plan to Current Month**.
4. In the **Analyse By** section, specify how to display the report.

*In AMT, **Equipment Plan** and **Equipment Terms** are used interchangeably.*

5. In the **Equipment Hierarchy** section, select:

    - Branch, Fleet, Site, Equipment and Model.
    - The relevant projection type:
        - **Current** will examine the current maintenance strategy.
        - Other projections will examine alternate scenarios that have been created.

6. In the **Additional Filters** section, add further details such as:

    - UOM (such as operating hours)
    - Cost Type
    - System
    - Cost Bearer
    - Sub System
    - Component Code
    - Modifier Code
    - Strategy Task
    - Escalation
    - Cost Selector

*All assets within the selected **Equipment Hierarchy** filters must have a common UOM to be able to run these reports.*

7. Click **Preview**.

8. The **Cost Per UOM - By Usage** report is displayed.

9. If you want to run the Cost per UOM - Comparison report, select the **Cost per UOM - Comparison** option from the **Cost per UOM** function.

10. Open Cost per UOM – Comparison report and Repeat Steps 3-7 to enter the relevant details of the asset and preview the report.

11. The **Cost Per UOM - Comparison** report is displayed. You can use the same procedure for accessing and running other benchmarking reports in AMT.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

2. Investigate Performance Drivers

Overview

Once the benchmarking reports provide information on the performance of an Equipment, you can investigate the drivers for the Equipment's good or poor performance. In this step, you review the relevant asset's life cycle projection's details (for example, Strategy Tasks and work order history) and check to see if they are in line with expectations.

Use the **Strategy Task Analyser** function in AMT to investigate performance drivers.

3. Strategy Optimisation

Overview

In this process, the assets' strategies and work order histories are compared to identify variances.

For more information, refer to the **Strategy Optimisation Training Manual.**

4. Economic Life Optimisation

Overview

Benchmarking report outcomes or Strategy Optimisation analysis may indicate that an equipment is performing poorly and needs to be disposed of.

Having located the poor performing equipment, in this step, we determine the optimal disposal point of the equipment.

For more information, refer to the **Economic Life Optimisation** section in this document.

5. Modelling

Overview

Having decided to terminate an equipment, it may need to be replaced with a new equipment.

New equipment options are modelled to replace the equipment that will be terminated, and life cycles of those equipment options are compared to identify the preferred potential replacement option.

For more information, refer to the **Modelling an Equipment Training Manual**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

6. Summary

Summary

The key points to remember from the Asset Benchmarking section of this guide are:

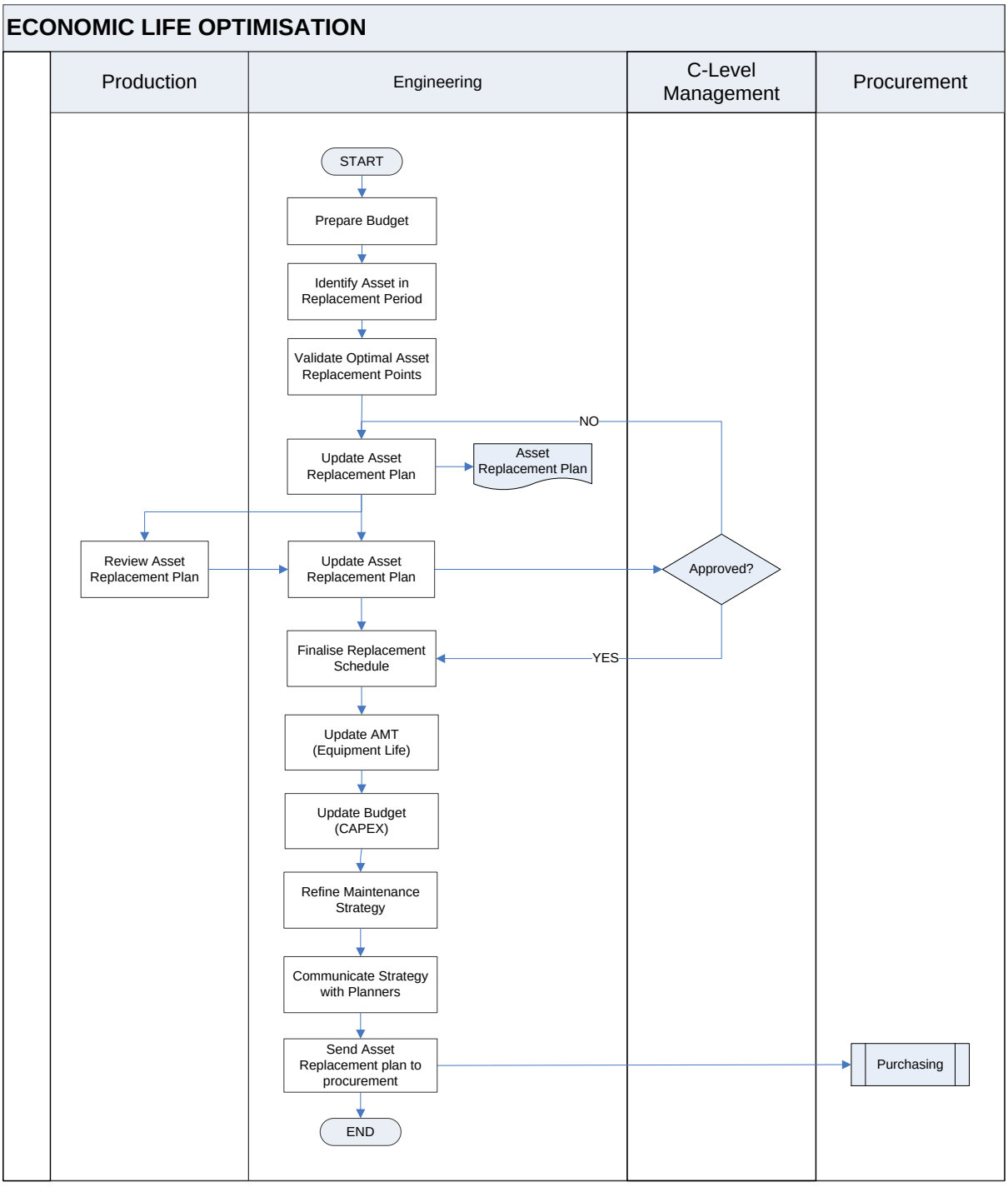
- Benchmarking is the process of comparing asset performance and maintenance strategies within a single site or across multiple sites.
- You can perform the following benchmarking steps in AMT:
    - Run Equipment benchmark reports.
    - Investigate performance drivers.
    - Strategy Optimisation.
    - Economic Life Optimisation.
    - Modelling.

Economic Life Optimisation

Overview

Economic Life Optimisation in AMT is the process of determining the optimal disposal point of an equipment.

Process Diagram



Training Workshops

7. Identify Equipment in replacement period

Overview

Identify potential equipment that need to be replaced in a specific period based on benchmark comparisons and other analysis.

Factors outside of AMT will be used to determine when to retire an asset, such as asset book value, new technologies in the market and new models released by an OEM.

8. Validate optimal Equipment replacement points

Overview

After identifying the Equipment, we determine the Equipment's optimal point of disposal using AMT's Residual Value in Components (RVIC) Chart report and RVIC Analysis report; as well as other inputs (for example, the asset's current book value).

There are two AMT reports that you can use to validate the optimal replacement points of an Equipment:

- RVIC Chart report
- RVIC Analysis report

RVIC Chart report

Procedure – RVIC Chart report

This report shows a line chart displaying the movement of residual life over the period of the chart dates.

Perform the following steps to run the RVIC Chart report:

1. Search for **RVIC - Chart** using the **Search** field and double-click **RVIC - Chart**



*Alternatively, from the **My Functions** hierarchy select **Reporting & Analyses** > **Equipment Evaluation** > **Residual Value Analyses** > **RVIC - Chart.***

2. On the **RVIC - Chart - Report Parameters** dialog, perform the following actions:
    - Enter the relevant details in the required fields.
    - Set the **Projection Type** as **Current**.
    - The **Cost Bearer** is **Contract** by default.
    - Click **Preview**.
3. The **Residual Value in Components** report is displayed.



This report shows the total value of components for an Equipment and does not drill down to its specific components.

RVIC Analysis report

Procedure – RVIC Analysis report

This report shows defined periods. Components can be used as a basis for contract variation or termination and can also be used for assessing contingent liabilities.

Perform the following steps to run the RVIC Analysis report:

1. Search for **RVIC Analysis** using the **Search** field and double-click **RVIC Analysis**.



*Alternatively, from the **My Functions** hierarchy select **Reporting & Analyses** > **Equipment Evaluation** > **Residual Value Analyses** > **RVIC Analysis.***

2. The AMT Standard Reports/RVIC Analysis Text - Report Parameters dialog is displayed. Perform the following actions:

   - Enter the relevant details in the required fields.
   - Set the **Projection Type** as **Current**.
   - The **Cost Bearer** is **Contract** by default.
   - Click **Preview**.

3. The **Residual Value in Components - Equipment** report is displayed.



Like the RVIC graphs, this report breaks down the RVIC by Equipment at a specific point in time as determined by the End Date when the report is run. While the RVIC graphs focus on how the Equipment changes over time, this report displays the forecast RVIC calculation at the selected End Date.

4. To view the task details, click the relevant Equipment Plan.
5. The Equipment's **RVIC Analysis Component - Detail** report is displayed.

9. Validate Equipment's Life Cycle Term

Overview

After determining the disposal and replacement point for an Equipment, we finalise the schedule and the life cycle term of the equipment.

Procedure – Validate Equipment's Life Cycle Term

Perform the following steps to validate an Equipment's life cycle term.

1. Search for **Equipment Administration** using the **Search** field and double-click **Equipment Administration**.



*Alternatively, from the **My Functions** hierarchy select **Setup & Admin** > **Equipment** > **Equipment Administration.***

2. The Equipment Administration dialog is displayed.

    - Select the relevant **Equipment**.
    - In the **Advanced Filters** section, set the relevant **Display Fields** such as **End Date** and **UOM**.
    - Click **Apply Filter**.

3. AMT displays a corresponding list.

4. Validate the **End Date**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

10. Refine Maintenance Strategy

Overview

The maintenance strategy for the Equipment in the context of the Asset's new life cycle term is refined. Changes to Maintenance Strategy such as Strategy Tasks, Non-planed Strategy Tasks and Unassigned Strategy Tasks impacts the long-term plan. Therefore, you also refine and update the long-term plan in the context of the Asset's new life cycle term.

For more information, refer to the **Strategy Optimisation Training Manual.**

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

11. Refine Long-Term Plan

Overview

A change in maintenance strategy impacts the long-term plan. Therefore, we also refine and update the long-term plan in the context of the Asset's new life cycle term.

For more information, refer to the **Long-Term Planning Training Manual.**

12. Summary

Summary

The key points to remember from this guide are:

- Economic Life Optimisation in AMT is the process of determining the optimal disposal point of an equipment.
- You can perform the following Economic Life Optimisation steps in AMT:
    - Identify Equipment in replacement period.
    - Validate optimal Equipment replacement points.
    - Validate Equipment's life cycle term.
    - Refine maintenance strategy.
    - Refine long-term plan.